Steps Followed Behind the Scenes (The Data Modelling and AI Insight)

Explained with GST as example, same applicability on CIT and SWT

Step 1: Importing Required Libraries

Explanation:

- **NumPy, Pandas, Matplotlib, Seaborn** are imported for data manipulation, visualisation, and statistical analysis.
- **Warnings are suppressed** to avoid clutter from non-critical alerts (e.g., future deprecations).
- **Pandas display settings** are adjusted to ensure all columns are visible during exploratory analysis.

Reason:

These libraries form the foundation for data preprocessing, visualisation, and model training.

Step 2: Loading the Dataset

Explanation:

- The dataset is loaded from an Excel file into a Pandas DataFrame.
- All the Data rows are displayed to verify data integrity (e.g., correct columns, no parsing errors).

Reason:

Ensures the dataset is loaded correctly before proceeding with analysis.

Step 3: Checking for Missing Values

Explanation:

- Null values are checked column-wise to identify gaps in the data.

Reason:

Missing values can distort model training. Identifying them early allows for proper handling (e.g., imputation or removal).

Step 4: Handling Missing Values
Explanation:

- Missing values in numerical columns (e.g., total_sales_income, gst_payable) are replaced with **0**.

Reason:

- Financial data often encodes "no transaction" as null. Replacing with 0 reflects business reality (e.g., no GST payable = ₹0).
- Prevents models from misinterpreting nulls as errors.

Step 5: Separating Segmented and Unsegmented Data
Explanation:

- Data is split into:
    - **Segmented**: Labelled records for training.
    - **Unsegmented**: Records marked "0-Unsegmented" for prediction.

Reason:
Segmented data trains the model; unsegmented data will be classified post-training.

Step 6: Checking Segmentation Distribution
Explanation:

- Determined the count of records per segment (e.g., “4-LTP”, “3-Medium”, ”2-Small” etc).

Reason:

- Detects class imbalance.
- Imbalance can bias the model toward the majority class, necessitating techniques like **SMOTE**.

Step 7: Selecting Features for Model Training
Explanation:

- Only numerical features (e.g., 'total_sales_income', 'exempt_sales', 'zero_rated_sales', 'gst_taxable_sales', 'deferred_import_liabilities', 'gst_paid_on_inputs','gst_paid_exempt_sales', 'gst_paid_private',) have been selected.
- Rest of the columns are not usable for the model.

Reason:

- Tree-based models (e.g., Random Forest) handle numerical data natively.
- Reduces noise from irrelevant features.

Step 8: Handling Class Imbalance with SMOTE
Explanation:

- **SMOTE** generates synthetic samples for minority classes (e.g., "Manufacturing") by interpolating existing data.

Reason:

- Without SMOTE, the model may ignore rare segments due to low representation.
- Unlike random oversampling, SMOTE avoids duplication, reducing overfitting.

Step 9: Splitting Data for Training and Testing
Explanation:

- Data is split into:
    - **Training (70%)**: Used to train the model.
    - **Testing (30%)**: Used to evaluate performance.

Reason:

- Ensures the model is tested on unseen data, validating its generalizability.

Step 10: Training the Random Forest Classifier

Explanation:

- **Random Forest** is chosen over alternatives (e.g., Neural Networks, Logistic Regression).

Reason:

1. **Handles Non-linearity**: Financial data often has complex relationships (e.g., total_sales_income vs. gst_payable).
2. **Robust to Outliers/Skewness**: Tree-based models split data hierarchically, making them less sensitive to skewed distributions (e.g., right-skewed gst_refundable).
3. **Feature Importance**: Provides interpretable insights into which features drive segmentation (e.g., gst_taxable_sales may be a key predictor).

Step 11: Evaluating the Model

Explanation:

- Metrics like **Accuracy, Precision, Recall, F1-score** are computed.

Step 12: Predicting Segmentation for Unsegmented Data

Explanation:

- The trained model predicts labels for unsegmented records.

Reason:

Automates manual classification, saving time and reducing human bias.

Step 13: Saving the Results

Explanation:

- Predictions are saved to a new CSV/Excel file.

Reason:

Enables downstream use in reporting or decision-making.

Key Insights from Additional Analyses

1. Correlation Matrix Interpretation

- **High Correlation (≥0.7)**:
    - total_sales_income ↔ zero_rated_sales (0.80): Expected, as zero-rated sales contribute to total income.
    - gst_paid_on_inputs ↔ gst_refundable (0.69): Refundable credits often arise from input taxes.
- **Action Taken**:
    - Retained all features despite correlations because:
        1. Random Forest inherently handles multicollinearity.
        2. Feature reduction risks losing nuanced patterns (e.g., gst_refundable may uniquely impact certain segments).

2. Histogram Interpretation

- **Right-Skewed Features**:
    - gst_payable, input_credits have most values near 0, with few high outliers.
- **Action Taken**:
    - No log transformation was applied because:
        1. Random Forest splits data based on values, not distributions.
        2. Outliers may represent legitimate high-value transactions (e.g., large refunds).

3. Zero Percentage Analysis

- **Sparse Features**:
    - gst_paid_private (95% zeros): Rarely used tax category.
- **Action Taken**:
    - Retained the column because even rare events may be critical for segmenting niche businesses.

Why Not Other Models?

- **Neural Networks**:
  - Require large data volumes and extensive tuning.
  - Overkill for structured tabular data with clear feature importance.
- **Logistic Regression**:
  - Assumes linear relationships, which financial data often violate.
- **Decision Trees**:
  - Prone to overfitting; Random Forest's ensemble approach mitigates this.

Conclusion

The workflow ensures robust segmentation by:

1. Addressing data quirks (sparsity, skewness) via Random Forest.
2. Balancing classes with SMOTE.
3. Validating feature relationships without aggressive reduction.
   The final model automates segmentation with high accuracy, enabling scalable business insights.

A. VALIDATION RULES FOR RAW GST DATA UPLOADS

Validation Rules will allow the ML to sweep through the raw data uploaded and separate the incomplete data for further investigation. The incomplete data needs to be completed and reuploaded for further ML screening and flag possible fraud or red flag detection. The whole process would be under the supervision of an officer higher in the organisational structure to oversee that no data gets out of the system undetected.

Rule 1: Validate TIN (Tax Identification Number)

- Must be exactly 9 digits (numeric) and First digit must not be 0.
- TIN must not be a sequence of consecutive numbers (e.g., 123456789).
- TIN must not consist of the same digit repeated (e.g., 555555555).

Rule 2: Validate Taxpayer Type

- Must not be null.
- Must be one of the allowed types:
  - "INDIVIDUAL" or "ENTERPRISE"

Rule 3: Validate Assessment Number

- Must be numeric.
- Must be unique (no duplicates allowed).
- Non-numeric and duplicate assessment numbers are removed.

Rule 4: Validate Tax Account Number

- Must be numeric.
- Non-numeric values are removed.

Rule 5: Validate Sales Columns

(Applies to: Total Sales Income, Exempt Sales, Zero-Rated Sales, Add Exempt and Zero-Rated Sales)

- Must be numeric values (integer or float) or must be non-negative (≥ 0).
- Invalid values are removed.

B. FLAG RULES FOR SUCCESSFULLY UPLOADED AND VALIDATED GST DATA

Flag Rules will allow the ML to sweep through the successfully uploaded data that has been successfully screened through validation check and have been found to be complete and validated. The ML will screen through the data to find possible misrepresentation and check for any requirement for further screening and audit. This will highlight the data which is found to be not match and have some possible rule break. The reasons will be provided by ML that it has found to be a mismatch and rule break for further investigation by Audit Officer.

The Red flags detected are marked in Red
The Possible reason for red flag detection is marked in blue.

Rule 1: Deducted Input Credits > Actual Input Credits

- Relationship: deduct_input_credits > input_credits
- Violation: Claimed more credits than available
- Red Flag: Potential fake invoices or duplicate entries
- Audit Risk: Verify all supplier invoices

Rule 2: Refund Claimed When Output Tax ≥ Deducted Credits

- Relationship: gst_refundable > 0 AND deduct_input_credits ≤ output_debits
- Violation: Claiming refund when tax liability exists
- Red Flag: Possible underreported sales
- Audit Risk: Cross-check sales records

Rule 3: Net Tax Due > 0 But Zero Payment

- Relationship: (output_debits - deduct_input_credits) > 0 AND gst_payable == 0
- Violation: Unpaid tax liability
- Red Flag: Avoiding tax payments
- Audit Risk: Payment reconciliation audit

Rule 4: Zero-Rated Sales > Total Sales

- Relationship: zero_rated_sales > total_sales_income
- Violation: Impossible export volume
- Red Flag: Fake export invoices
- Audit Risk: Verify export documentation

Rule 5: Current Taxable Sales < 50% Previous Period With Same/More Input Credits

- Relationship: gst_taxable_sales < (0.5 × gst_taxable_sales_prev) AND input_credits ≥ input_credits_prev
- Violation: Sales drop with maintained expenses
- Red Flag: Moving sales to exempt category
- Audit Risk: Analyze sales classification

Rule 6: Six+ Consecutive Refund Months with Sales < PGK 250,000

- Relationship: consecutive_refund_months ≥ 6 AND total_sales_income < 250000
- Violation: Minimal operations but regular refunds
- Red Flag: Shell company activity
- Audit Risk: Supplier transaction tracing

Rule 7: Exempt Sales > 50% Total Sales

- Relationship: exempt_sales > (0.5 × total_sales_income)
- Violation: Excessive exempt transactions
- Red Flag: Possible misclassification
- Audit Risk: Review exempt sales documentation

Rule 8: GST on Inputs > 30% Total Sales

- Relationship: gst_paid_on_inputs > (0.3 × total_sales_income)
- Violation: Disproportionate input tax
- Red Flag: Inflated purchase claims
- Audit Risk: Supplier invoice verification required

Rule 9: Output Tax < 5% Total Sales

- Relationship: output_debits < (0.05 × total_sales_income)
- Violation: Suspiciously low output tax
- Red Flag: Underreported taxable sales
- Audit Risk: Sales vs bank deposit comparison

Rule 10: GST Refund > 20% Total Sales

- Relationship: gst_refundable > (0.2 × total_sales_income)
- Violation: Excessive refund claims
- Red Flag: Potential circular trading
- Audit Risk: Refund claim investigation required

Rule 11: Zero-Rated + Exempt Sales > Total Sales

- Relationship: (zero_rated_sales + exempt_sales) > total_sales_income
- Violation: Impossible tax-free volume
- Red Flag: Fabricated exempt transactions
- Audit Risk: Full exempt sales audit

Rule 12: Deferred Import Liabilities > 50% Total Sales

- Relationship: deferred_import_liabilities > (0.5 × total_sales_income)
- Violation: Excessive deferral claims
- Red Flag: Improper import tax handling
- Audit Risk: Import documentation review

Rule 13: Private Use GST > 10% Total Input GST

- Relationship: gst_paid_private > (0.1 × gst_paid_on_inputs)
- Violation: High personal expense claims

- Red Flag: Business/personal expense mixing
- Audit Risk: Expense categorization check

Rule 14: Negative Taxable Sales

- Relationship: gst_taxable_sales < 0
- Violation: Impossible negative sales
- Red Flag: Accounting manipulation
- Audit Risk: Complete sales records audit

Rule 15: Sudden Spike in Sales Without Matching Input Credits

- Relationship: (sales reported in one quarter > 150% of previous quarter) and (input_credits claimed ≤ previous quarter)
- Violation: Unexplained surge in sales with stagnant or reduced input credits
- Red Flag: Manipulated sales figures to inflate GST liabilities or obtain undue refunds
- Audit Risk: Review of sales and input credit reconciliation across quarters

Rule 16: High Refund Claims with Low Sales

- Relationship: total GST refunds claimed in a quarter > PGK 100,000) and (minimal declared sales 10%
- Violation: Disproportionate GST refund compared to sales
- Red Flag: Possible fictitious transactions or fake invoices
- Audit Risk: Full audit of refund claims and corresponding sales records

Rule 17: Fewer than 12 Tax Months Reported Per Year

- Relationship: count(tax_period_month) < 12 per (TIN, tax_period_year)
- Violation: Incomplete monthly tax filings
- Red Flag: Possible non-compliance or inactive months
- Audit Risk: Filing frequency and business activity consistency check

Rule 18: No Tax Months Reported in a Year

- Relationship: count(tax_period_month) = 0 per (TIN, tax_period_year)

- Violation: Missing annual tax returns
- Red Flag: Possible tax evasion or dormant business
- Audit Risk: Full-year non-filing investigation

C. VALIDATION RULES FOR RAW SWT DATA

Validation Rules will allow the ML to sweep through the raw data uploaded and separate the incomplete data for further investigation. The incomplete data needs to be completed and reuploaded for further ML screening and flag possible fraud or red flag detection. The whole process would be under the supervision of an officer higher in the organisational structure to oversee that no data gets out of the system undetected.

Rule 1: Validate TIN (Tax Identification Number)

- Must be exactly 9 digits (numeric).
- First digit must not be 0.
- Must not be a sequence of consecutive numbers (e.g., 123456789).
- Must not consist of the same digit repeated (e.g., 555555555)

Rule 2: Validate Employees Paid SWT vs Employees on Payroll

- Employees Paid SWT cannot exceed Employees on Payroll.
- Violations suggest incorrect SWT reporting or payroll data.
  →Records with Invalid records are removed.

Rule 3: Validate Head Office Column

- Must contain only 'Y' (Yes) or 'N' (No).
- Blank or NA values are allowed.
  → Records with Any other values are removed.

Rule 4: Validate Tax Period

- Month must be between 1 and 12.
- Year must be between 2010 and the current year.

→ Records with Invalid periods are removed.

Rule 5: Validate Entry Date

- Date must not be in the future.
- Must follow a valid date format (e.g., YYYY-MM-DD).

  →Records with Future dates and invalid date formats are removed.

Rule 6: Validate Assessment Number

- Must be unique (no duplicates).
- First occurrence is retained; subsequent duplicates are removed.
- Blank or NA values are allowed.

Rule 7: Validate Numeric Columns

Negative values are not allowed in the following fields:

- Employees on Payroll
- Total Salary Wages Paid
- Employees Paid SWT
- SW Paid for SWT Deduction
- Total SWT Tax Deducted

  → Records with Negative values are removed.

Rule 8: Validate Integer Fields

Fractional values are not allowed in the following fields:

- Employees on Payroll
- Employees Paid SWT

  → Records with Fractional values are removed.

Rule 9: Validate Date Columns Format

The following date columns must have valid date formats:

- Entry Date
- Assessed Date
- Due Date
  → Records with invalid date formats are removed.

D. FLAG RULES FOR SUCCESSFULLY UPLOADED AND VALIDATED SWT DATA

Flag Rules will allow the ML to sweep through the successfully uploaded data that has been successfully screened through validation check and have been found to be complete and validated. The ML will screen through the data to find possible misrepresentation and check for any requirement for further screening and audit. This will highlight the data which is found to be not match and have some possible rule break. The reasons will be provided by ML that it has found to be a mismatch and rule break for further investigation by Audit Officer.

The Red flags detected are marked in Red
The Possible reason for red flag detection is marked in blue.

Rule 1: SWT Declared, But No Tax Deducted
Condition: employees_paid_swt > 0 and total_swt_tax_deducted == 0
Violation: Employees are paid, but no tax is withheld
Red Flag: Tax avoidance or payroll non-compliance
Audit Risk: High – Indicates deliberate evasion or system errors

Rule 2: SWT Deductions Exceed Total Salary Paid
Condition: sw_paid_for_swt_deduction > total_salary_wages_paid
Violation: Deductions exceed salary
Red Flag: Fake payroll entries or inflated tax claims
Audit Risk: High – Strong indication of fraudulent reporting

Rule 3: Tax Deducted Exceeds SWT-Eligible Salary
Condition: total_swt_tax_deducted > sw_paid_for_swt_deduction

Violation: Excessive tax withheld
Red Flag: Incorrect tax calculations or manipulated entries
Audit Risk: Medium – Requires recalculation and verification

Rule 4: More Taxed Employees Than on Payroll
Condition: employees_paid_swt > employees_on_payroll
Violation: Taxed employees exceed payroll
Red Flag: Ghost employees
Audit Risk: High – Suggests systemic payroll fraud

Rule 5: Employees on Payroll with Zero Salary
Condition: total_salary_wages_paid == 0 and employees_on_payroll > 0
Violation: No wages paid despite listed staff
Red Flag: Off-the-books payments or underreporting
Audit Risk: High – Likely wage evasion

Rule 6: Deductions With No Eligible Employees
Condition: employees_paid_swt == 0 and sw_paid_for_swt_deduction > 0
Violation: Deductions without eligible staff
Red Flag: Fictitious deductions
Audit Risk: High – Indicates record manipulation

Rule 7: Tax Deducted Without SWT Payments
Condition: sw_paid_for_swt_deduction == 0 and total_swt_tax_deducted > 0
Violation: Deducted tax but no salary basis
Red Flag: Inflated deductible amounts
Audit Risk: High – Points to tampering or fabrication

Rule 8: Employees Taxed but Not on Payroll

Condition: employees_on_payroll == 0 and employees_paid_swt > 0

Violation: Deductions for non-existent staff

Red Flag: Ghost payroll

Audit Risk: High – Requires employee verification

Rule 9: Salaries Paid but No Employees Listed

Condition: total_salary_wages_paid > 0 and employees_on_payroll == 0

Violation: Salary payments without employees

Red Flag: Fabricated payrolls

Audit Risk: High – Suspected money laundering or tax evasion

Rule 10: Excessive Difference in SWT vs Payroll Employees

Condition: employees_paid_swt > employees_on_payroll * 1.1

Violation: >10% difference

Red Flag: Duplicate/fake employee records

Audit Risk: Medium – Requires deeper payroll audit

Rule 11: Deductions Exceed Total Salary

Condition: (sw_paid_for_swt_deduction / total_salary_wages_paid) > 1

Violation: Deductions > 100% of salary

Red Flag: Blatant fraud or data issues

Audit Risk: Very High – Immediate investigation

Rule 12: Salary Spike Without Payroll Growth

Condition: YOY change % of total_salary_wages_paid > 30% and YOY change % of employees_on_payroll < 10%

Violation: Salary changes not supported by workforce size

Red Flag: Temporary salary inflation for higher deductions

Audit Risk: Medium – Requires historical trend review

E. VALIDATION RULES FOR UPLOADED RAW CIT DATA

Validation Rules will allow the ML to sweep through the raw data uploaded and separate the incomplete data for further investigation. The incomplete data needs to be completed and reuploaded for further ML screening and flag possible fraud or red flag detection. The whole process would be under the supervision of an officer higher in the organisational structure to oversee that no data gets out of the system undetected.

Rule 1: TIN (Taxpayer Identification Number) Validation

- Must be exactly 9 digits (^\d{9}$)
- Cannot start with zero
- Cannot have all identical digits (e.g., 111111111)
- Cannot be a continuous sequence (e.g., 123456789, 987654321)
- Cannot be null or empty

Rule 2: Assessment Number Validation

- Must contain only digits (numeric)
- No duplicate values allowed

Rule 3: Tax Account Number Validation

- Must be numeric (only digits)

Rule 4. Gross Sales Validation

- Must be numeric
- Cannot be negative
- Cannot exceed Total Gross Income by more than 1%

Rule 5. Rental Expenses and Stamp Duty Check

- If 126.Rental expenses > 0, then check for corresponding Stamp Duty payments

F. FLAG RULES FOR SUCCESFULLY UPLOADED AND VALIDATED CIT DATA

Flag Rules will allow the ML to sweep through the successfully uploaded data that has been successfully screened through validation check and have been found to be complete and validated. The ML will screen through the data to find possible misrepresentation and check for any requirement for further screening and audit. This will highlight the data which is found to be not match and have some possible rule break. The reasons will be provided by ML that it has found to be a mismatch and rule break for further investigation by Audit Officer.

The Red flags detected are marked in Red
The Possible reason for red flag detection is marked in blue.

Rule 1: Gross Sales Validation

- Condition: 10. Gross_sales_(cash_/_credit) < 90% of 90. TOTAL_GROSS_INCOME
- Violation: Understated gross sales relative to income
- Red Flag: Suspected sales concealment
- Audit Risk: Income underreporting audit

Rule 2: Cost of Goods Sold Validation

- Condition: 100.Cost_of_goods_sold > 75% of 10. Gross_sales_(cash_/_credit)
- Violation: Cost of goods sold exceeds sales
- Red Flag: Inflated expense reporting
- Audit Risk: Profit margin manipulation audit

Rule 3: Property & Leasehold Validation

- Condition: 520.Property_/_Equipment < 521.Leasehold_Improvements
- Violation: Leasehold exceeds total property value
- Red Flag: Asset reporting inconsistency
- Audit Risk: Fixed asset audit

Rule 4: Foreign Management Fees Validation

- Condition: 123.Management_fees_-_Foreign > 20% of 190.TOTAL_OPERATING_EXPENSES
- Violation: Excessive foreign management fees
- Red Flag: Base erosion risk
- Audit Risk: Related party transactions audit

Rule 5: Foreign Royalties Validation

- Condition: 128.Royalties_-_Foreign > 15% of 90. TOTAL_GROSS_INCOME
- Violation: Excessive foreign royalty payments
- Red Flag: Intellectual property mispricing
- Audit Risk: Royalty abuse audit

Rule 6: Advertising & Promotion Validation (Operating Expenses)

- Condition: 104.Advertising_and_promotion > 10% of 190.TOTAL_OPERATING_EXPENSES
- Violation: Unusual promotion spends
- Red Flag: Expense inflation
- Audit Risk: Expense authenticity audit

Rule 7: Bad Debts Validation

- Condition: 105.Bad_debts_written_off > 5% of 502.Accounts_receivable_(trade)
- Violation: High level of bad debts
- Red Flag: Possible revenue manipulation
- Audit Risk: Receivables audit

Rule 8: Commissions Validation

- Condition: 107.Commissions > 15% of 10. Gross_sales_(cash_/_credit)
- Violation: Excessive commission expenses
- Red Flag: Inflated payouts
- Audit Risk: Agent/related party audit

Rule 9: Consultancy Fees Validation

- Condition: 109.Consultancy_fees > 10% of 190.TOTAL_OPERATING_EXPENSES
- Violation: Excessive consultancy charges
- Red Flag: Cost inflation risk
- Audit Risk: Third-party service audit

Rule 10: Legal Expenses Validation

- Condition: 121.Legal_expenses > 5% of 190.TOTAL_OPERATING_EXPENSES
- Violation: Unusual legal expenditure
- Red Flag: Potential hidden settlements
- Audit Risk: Legal services audit

Rule 11: Repairs & Maintenance Validation

- Condition: 125.Repairs_and_maintenance > 8% of 520.Property_/_Equipment
- Violation: High maintenance relative to assets
- Red Flag: Possible misclassification of capital expenses
- Audit Risk: CapEx vs OpEx audit

Rule 12: Travel & Accommodation Validation

- Condition: 132.Travel_and_accommodation > 7% of 190.TOTAL_OPERATING_EXPENSES
- Violation: Excessive travel claims
- Red Flag: Possible misuse of business funds
- Audit Risk: Expense audit

Rule 13: Other Gross Income Validation

- Condition: 20. Other_gross_income < 5% of 90. TOTAL_GROSS_INCOME
- Violation: Understated other income
- Red Flag: Concealed income streams

- Audit Risk: Revenue source audit

Rule 14: Other Current Assets Validation

- **Condition: 504.Other > 20% of 515.TOTAL_CURRENT_ASSETS**
- **Violation: Excessive unclassified current assets**
- Red Flag: Potential misreporting or asset hiding
- Audit Risk: Balance sheet audit

Rule 15: Gross Sales vs Total Gross Income Validation

- Condition: 10. Gross_sales_(cash_/_credit) > 90. TOTAL_GROSS_INCOME
- Violation: Sales exceed declared gross income
- Red Flag: Revenue reporting inconsistency
- Audit Risk: Income reconciliation audit

Rule 16: Foreign Interest Expense Validation

- Condition: 119.Interest_expense_-_Foreign > 17. Interest_income
- Violation: Foreign interest exceeds income
- Red Flag: Thin capitalization risk
- Audit Risk: Interest deduction abuse audit

Rule 17: Foreign vs PNG Management Fees Validation

- Condition: 123.Management_fees_-Foreign > 122.Management_fees-_PNG
- Violation: More fees paid abroad than locally
- Red Flag: Profit shifting
- Audit Risk: Transfer pricing audit

Rule 18: Foreign vs PNG Royalties Validation

- Condition: 128.Royalties_-Foreign > 127.Royalties-_PNG
- Violation: Foreign royalties exceed domestic

- Red Flag: IP base erosion
- Audit Risk: IP structuring audit

Rule 19: Advertising & Promotion vs Gross Sales Validation

- Condition: 104.Advertising_and_promotion > 10% of 10. Gross_sales_(cash_/_credit)
- Violation: Over-promoting relative to sales
- Red Flag: Inefficient spending or laundering
- Audit Risk: Cost-effectiveness audit

Rule 20: Depreciation vs Property & Equipment Validation

- Condition: 111.Depreciation > 25% of 520.Property_/_Equipment
- Violation: Depreciation exceeds standard threshold
- Red Flag: Possible overstatement of asset wear
- Audit Risk: Asset value manipulation audit

Rule 21: Dividend Income Validation

- Condition: 15. Dividend_income > 20% of 90. TOTAL_GROSS_INCOME
- Violation: Unusual level of dividend income
- Red Flag: Income misclassification
- Audit Risk: Passive income audit

Rule 22: Interest Expense vs Loans Validation

- Condition: (118.Interest_expense_-PNG + 119.Interest_expense-_Foreign) > 15% of (561.Loans_from_directors + 562.Other_loans)
- Violation: Interest expense exceeds allowable ratio
- Red Flag: Thin capitalization risk
- Audit Risk: Debt structure audit

Rule 23: Gross Income Composition Validation

- Condition: (90. TOTAL_GROSS_INCOME - 10. Gross_sales_(cash_/_credit)) > 25% of 90.TOTAL_GROSS_INCOME

- Violation: Other income disproportionately high
- Red Flag: Potential misstatement of revenue sources
- Audit Risk: Revenue categorization audit

Rule 24: Bad Debt Written Off Validation

- Condition: 105.Bad_debts_written_off > 1% of 502.Accounts_receivable_(trade)
- Violation: Bad debts unusually high
- Red Flag: Suspected financial misrepresentation
- Audit Risk: Credit policy and receivables audit

Rule 25: Prior Year Losses Validation

- Condition: 308.Prior_Year_Losses_Utilised ≠ 0
- Violation: Prior year losses claimed
- Red Flag: Potential carry forward misuse
- Audit Risk: Loss utilization audit

Rule 26: Tax Calculation Consistency Validation

- Condition: (90. TOTAL_GROSS_INCOME - 190.TOTAL_OPERATING_EXPENSES + 290.TOTAL_NON_DEDUCTIBLE_ITEMS - 390.TOTAL_DEDUCTIBLE_ITEMS) × 30% ≠ 720.GROSS_TAX
- Violation: Inconsistency in tax calculation
- Red Flag: Possible computation or reporting error
- Audit Risk: Tax liability under/overstatement audit.

Rule 27: Match GST Sales with CIT Gross sales

- Condition: Matching of Total 12 months period of GST sales from GST Returns against CIT Sales Declared, given total sales > PGK 250,000
- Violation: Understatement or Overstatement of Sales with GST misinformation.
- Red Flag: Sales suppress or GST non-payment
- Audit Risk: Check for matching CIT and GST Return

Rule 28: Match SWT Salary and Wages with CIT Salary and Wages

- Condition: Match SWT returns of 12 months period with CIT salaries and wages
- Violation: Understatement or Overstatement of Salary and Wages with SWT misinformation.
- Red Flag: Expense overcharged or SWT non-payment
- Audit Risk: Check for matching CIT and SWT Return

RISK MODELING

The model uses a Random Forest Classification algorithm to detect potentially fraudulent records by calculating the flag percentage through the predict_proba() function, which estimates the likelihood (ranging from 0% to 100%) of a record being fraudulent. If this predicted probability exceeds 50%, the model identifies the record as a potential risk by setting is_flag = 1. For these flagged cases, the flag_percentage reflects the model's confidence level — higher values (e.g., 90–100%) indicate stronger fraud signals. The flag_logic column captures the specific rule violations (such as "Overclaimed Credits") that triggered the flag, providing transparency into the reasoning behind each prediction.

Risk types are determined based on predefined thresholds: "Very Very High Risk" (100%), "Very High Risk" (95–99%), "High Risk" (90–94%), "Medium Risk" (75–89%), "Low Risk" (60–74%), and "Very Low Risk" (50–59%). These thresholds enable a structured and systematic risk assessment process, ensuring that higher-risk cases receive stricter scrutiny during audits. The risk type logic helps prioritize flagged cases according to severity, while the binary is_flag variable acts as an initial filter, marking only those records where the model predicts more than 50% probability of fraud.

Please Note while special care has been taken by our financial analyst and financial fraud detection team to find out the best of the rules that detects a possible need for further investigation, but with the minimum level of data made available to us, we have tried to focus on balancing the possible advanced level detection through the help of ML. As more and more forms of structured and unstructured data from multiple levels and sources will be made available to us, we would be able to train the ML model to detect more advanced and complex matrix.